

Database Migration Guide

Applicable Roles: DEVELOPER, DEVOPS

Migration Principles

- Always version-controlled (Alembic/Flyway)
- One migration = one logical change
- Always testable and reversible
- Never edit existing migrations
- Review by DBA or senior dev

Safe Migration Techniques

- Expand-contract pattern for schema changes
- Add new column first, deploy code, remove old column
- Use background migrations for large tables
- Run EXPLAIN ANALYZE before and after
- Set statement timeout to prevent long locks

Rollback Strategy

Every migration must have a downgrade path:

- Complex migrations: test rollback
- Data migrations: backup before running
- Schema changes: reversible operations
- Have a rollback window (max 30 min)